

Harris Adnan

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EDUCATION

University of Houston

Master of Science in Computer Science

Houston, TX

Aug 2025 – May 2027

University of Houston

Bachelor of Science in Computer Science | GPA: 3.6/4.0 | Dean's List (2023–2024)

Houston, TX

Aug 2022 – Dec 2024

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, PHP, HTML, CSS, Bash

Frontend: React, Next.js, Vue.js, Nuxt.js, Tailwind CSS, Responsive Design, Component Architecture

Backend: Node.js, Express, REST APIs, PHP, Server-Side Rendering, Authentication

Databases: PostgreSQL, MySQL, ChromaDB, Database Design, SQL Optimization, pgAdmin

Developer Tools: Git, GitHub, Linux, MacOS, Jupyter, VS Code, Postman, npm, Agile/Scrum

Concepts: Data Structures, Algorithms, OOP, Full-Stack Development, System Design, Software Testing, CI/CD, API Integration, Vector Databases, LLM Integration

PROJECTS

TaskIQ | React, Node.js, REST APIs, Google Gemini API | taskiq.dev

Nov 2025 – Present

- Founded AI automation startup and shipped production full-stack web application serving service businesses across Med Spa, Home Services, and Tutoring verticals; reduced lead response time from industry-standard 48–72 hours to 3–4 seconds (~50,000x improvement).
- Architected scalable React frontend and Node.js backend with REST API endpoints for real-time message ingestion, lead scoring (0–100), and AI response generation; deployed to production with end-to-end error handling.
- Engineered revenue-at-risk calculation engine combining vertical-specific ticket values, attach-rate multipliers, and AI-derived confidence scores to quantify dollar impact of ignored leads in real time.
- Integrated Google Gemini 2.5 Flash API with structured JSON output schemas for intent classification, urgency detection, and brand-voice-matched reply drafting; engineered prompt pipelines tested across hundreds of inquiry variants.

Aircraft Trajectory Optimization | Python, NumPy, SciPy, CasADi, IPOPT

Feb 2026

- Implemented and compared Single Shooting vs. Direct Collocation for nonlinear aircraft trajectory optimization using CasADi and IPOPT, leveraging automatic differentiation for solver accuracy.
- Achieved 127.10 kg minimum fuel burn vs. 130.48 kg baseline (2.6% improvement) and validated robustness of Direct Collocation under failed Single Shooting conditions and mesh refinement tradeoffs.

CoogEsports Website | Nuxt.js, Vue.js, Tailwind CSS, PostgreSQL, REST APIs

Dec 2024

- Developed full-stack features for high-traffic student organization website serving 2,100+ event attendees and 1,000+ active members; built reusable Vue components and server-rendered pages with Nuxt.js.
- Reduced page-load time to ~1 second through server-side rendering and optimized PostgreSQL queries; implemented responsive design patterns for mobile and desktop users.

EXPERIENCE

University of Houston — Summer Research Fellowship

Houston, TX

Research Assistant, Dr. Weihang Zhu

May 2024 – Aug 2024

- Awarded competitive \$5,100 fellowship to architect and ship full-stack AI application analyzing 2,000+ offshore crane incident reports using Python, PostgreSQL, ChromaDB, and large language models.
- Built end-to-end safety-query system integrating PHP web frontend with Streamlit chatbot and PostgreSQL document store, enabling user authentication, document access, and keyword search across the full incident corpus.
- Designed ChromaDB-backed vector retrieval pipeline with LLM query handling, replacing manual document review with natural-language Q&A for root cause and remediation pattern extraction.

Code Ninjas

Houston, TX

Coding Instructor

Jan 2024 – May 2024

- Taught Python, JavaScript, and game development to 25+ students aged 8–16 through project-based curriculum covering algorithms, debugging, and software design.

LEADERSHIP

CougarCS, University of Houston

Houston, TX

Director of Tutoring (Incoming)

Aug 2026 – Present

- Selected to lead tutoring branch for the largest CS student organization at the University of Houston, building a consistent support system for CS majors across core coursework.
- Coordinating tutors and workshop directors to design review sessions, workshops, and learning resources that scale individual support into reusable materials.

Coog Esports, University of Houston

Houston, TX

President

Jan 2024 – Dec 2024

- Led largest student organization on campus (1,000+ members, 15+ officers); organized CougarCon flagship event with 2,100 attendees; secured corporate partnerships with Red Bull, Monster, and TDECU.